MATHMAG BUILDERS LIMITED

MANAGEMENT SYSTEM

Master System Architecture & Governance

Seven core modules • One operational platform • One authority model

Plan better, build better.

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Status: Business requirements harmonized • Developer implementation not yet performed

00. Executive decision and controlled-document precedence

The Mathmag Builders Management System (MBMS) contains seven CORE MODULES. Modules 1-6 are the client/project delivery lifecycle. Module 7 - Central Administration - is cross-cutting and operates continuously across them. Client, employee, subcontractor, supplier and management dashboards are role-specific portals within this single platform, not extra business modules.

This V3 document and the corresponding module V3 addenda are the newest governance/integration layer. They preserve previously approved module requirements except where an older generic or inconsistent statement must be interpreted using the later owner vs GM, delegated approval, independent review, statutory/safety and audit policies. Legal compliance and monetary delegation limits need verification and configuration; no software implementation is claimed.

Document	Priority and purpose
Signed contracts and law	Mandatory applicable legislation and actual contract obligations are binding; developer must not infer nonwaivable waivers.
Specific later user-confirmed rules	Higher priority than older illustrative drafting assumptions, especially safeguards and escalation.
Module 7 and this V3 architecture	Company-wide governance, HR, compliance, finance, identity, audit and permission rules.
Module 1-6 V3 addenda	Controlling business-rule interpretation of the attached preserved original module text.
Preserved source module pages	Authoritative detailed workflow where no conflict with newer rules.
Developer design proposals	Implementation suggestions only, subject to review, tests and approval.

01. Seven-module business architecture

Module	Purpose	Entry / exit
01 • Site Visit	Client intake after agreed visit date; mobile site survey and report.	Agreed booking -> reviewed published site brief.
02 • Planning	2D approval -> 3D -> Structural -> Mechanical/Plumbing -> Electrical -> QS.	Versioned approved technical package and budget.
03 • Client Acceptance	Management-published technical/budget proposal; exact-version acceptance; verified finance gate.	Authorized release to Module 4 or contracted permitted route.
04 • Approvals	Controlled printing, professional certification and statutory authority tracking.	Conditional planning completion; mandatory construction requirements separately cleared.
05 • Contracts	Service selection, negotiation, final agreement, execution, project appointment.	Construction planning-only activated if signed; physical work still gated.
06 • Construction	Procurement, workers, sites, costs, safety, handover, warranty.	Formal project closeout after all distinct obligations.
07 • Administration	HR, Accounts/Finance, Procurement administration, Compliance, QA, independent audit, corporate IT.	Continuous, company-wide; no sequential project gate.

Service pathways are progressive and conditional. Not every client buys construction: planning-only, printing-only, approval-only, technical inspection/supervision, labour-only and full-build engagements receive only relevant permitted scope. No system rule should force all modules as identical paid steps.

02. Organization and authority

Layer	Authority and restriction
Business Owner	Company ownership and strategic decisions; approved GM appointment/authority mandate, governance oversight, major reserved transactions, external audit/independent cases; not daily site/HR/procurement approval.
General Manager (GM)	Company-wide operational authority; sets/delegates functional matrices, settles GM-level escalations, controls company operating policy, monitors performance and risk.
Department Head	Technical, contract, HR, finance, procurement, compliance or QA delegated action within project/category/value/risk permissions.
Project Manager (PM)	Execution and site oversight for assigned project; requests resources, verifies progress, recommends major decisions; has only specifically delegated monetary authority.
Site Supervisor / Storekeeper / Worker	Executes distinct worker duties; originals never suppressed; cannot certify own independent approval.
Independent Auditor	Protected access to immutable records and protected complaint channel; appointments/removal and GM-related case routing must be insulated from conflicted management.
Qualified professionals	Independent scope-limited certified technical opinions/signatures under their validated professional eligibility.
Clients and suppliers	Limited own record access; acceptance and supplier banking verified independently; no cross-project information leakage.

02.1 Delegation and approval interpretation

- Every power has authority_category, role/person grant, project scope, value threshold, risk category, effective dates, evidence requirement and substitution/escalation route.
- “Management” in existing Module 1-6 documents means a valid delegated manager for ordinary decisions and the GM for above-limit or expressly reserved operational approvals; it never implicitly means owner approval.
- The owner sees protected strategic/audit risk summaries but remains outside routine finance, supplier, attendance, project deadlines and contract-related operational queues unless the matter is expressly owner-reserved.
- Mandatory technical, client, legal, HR and finance review steps are not optional just because a GM has spending authority.
- PM responsibility to coordinate is distinct from authority to contract, sign, make payment, change material ownership, certify a structural drawing or approve an independent investigation.
- GM absence: HR activates a time-limited preapproved acting deputy according to standing approved succession; if no deputy is available, preapproved emergency management committee operates via limited quorum and recorded votes, never exercising owner-reserved powers.

02.2 Cross-functional decision checkpoints

Case	Originator	Independent verification	Final authorization
Site cost/visit	Desk/site lead	Accounts amount and project; supervisor activity	Assigned delegated manager/GM on escalation
Design professional signoff	Authorized designer	Department technical lead/required professional	Qualified signatory; not finance/owner substitution
Proposal release	Lead QS	QS costing + Contracts where necessary + Accounts gate	Eligible delegated manager/GM
Permit application/fee	Approvals Officer	Professional + Compliance	Authorized manager for

Case	Originator	Independent verification	Final authorization
		applicability; Accounts payments	spend/publication
Binding client contract	Contracts Officer	Contracts/QS/Accounts/ professional scope checks	Authorized company signatory per actual mandate
Material PO	PM / site requisitioner	Procurement quotes + QS value + Accounts funding + technical review	Delegated purchasing approver/GM
Supplier payment	Accounts payable preparation	PO, accepted GRN, invoice; verified payment destination	Independent payment authorizer; separate payer/reconciler
Payroll	HR/PM verified input	Accounts calculation and statutory checks	Delegated management approval; Accounts payout
Critical safety return-to-service	Operator/worker original report	Qualified independent technical/safety assessment	Competent documented release and extra GM/PM release as policy requires
Final handover	PM	Qualified technical, QS final account, Accounts and Contracts	Delegated management formal release; separate client acceptance

03. Roles, portals and access model

Use role-based access PLUS project assignment, relationship type, document classification, service entitlements and record ownership. One person may hold multiple approved roles but the backend must block self-approval on conflicted transactions.

Portal	Must display	Must NOT display by default
Owner	Company KPIs, material exposure, GM reports, strategic exceptions, independent audit escalations.	Routine employee payroll details, all unfiltered client files and day-to-day approval chores.
GM	Operational summary, consolidated department queues, exceptions, critical site/finance/compliance and staffing risks.	Restricted whistleblower identity on conflicted case without authorization.
Department Head	Owned cases, assigned work, due dates, professional checks, permitted document revisions.	Unrelated employee grievances, unassigned client data, other confidential suppliers.
PM	Assigned project worker originals, site verification, procurement, schedule, QA and cost snapshots.	Secret audit informant identity, unassigned project ledgers.
Worker	Own tasks, original submissions, disputes, attendance, equipment/PPE acknowledgments, lawful own earnings.	Other workers wages/credentials, sensitive commercial quotes and audits.
Subcontractor	Own contracted scope, personnel assignments, claims, retention, evidence and handover.	Mathmag direct employment payroll or another subcontractor files.
Client	Published project drawings, approvals, agreed budgets, acceptance, invoices, selected progress updates, handover/warranty.	Editable CAD, internal incidents, raw employee media not published and audit results.
Independent auditor	Allowed protected originals, append-only transaction audit, investigations, corrective action verification.	Changing primary transactional records or deciding operational payments.

04. Project lifecycle and release gates

Gate	Required proof	Prohibited bypass
G1 / visit activated	Site date/time mutually agreed and project record opened.	Unqualified enquiry converted automatically to project.
G2 / architectural 3D	Client accepted specific 2D file revision.	3D production or claims based on informal outdated drawings.
G3 / engineering chain	Department lead approval for Architectural -> Structural -> MEP -> Electrical -> QS.	Unreviewed source file passing to next stage.
G4 / client proposal	Lead QS technical approval, authorized management publication, exact acceptance version.	Client approval silently changing when budget/drawings revised.
G5 / approvals stage	Client acceptance verified, financial milestone satisfied or explicit eligible finance exception and separate Module 4 release.	Financial bypass alters ledger or erases outstanding debt.
G6 / contract service activation	Management-approved executed final contract, signed evidence, service entitlements, appointed PM where appropriate.	Client service selection treated as signed construction authorization.
G7 / mobilization	Required permits and NCA, verified milestone/exception, PM documented readiness and management physical-start release.	Construction simply because M4 conditionally closed or M5 was signed.
G8 / critical work hold	Relevant competent stage/safety professional and operational release as applicable.	GM financial override replacing technical approval.
G9 / variation work	Scope/technical feasibility, QS rate/schedule, Contracts document, manager offer and client exact acceptance.	Chargeable extra work authorized solely by supervisor.
G10 / completion	Technical certificates, PM completion, QS/Accounts reconciliation, management handover, client evidence.	Single handover flag clearing unresolved financial or warranty duty.

05. Shared record architecture

Canonical identifiers and append-only source/evidence are the foundations. A single project reference follows all modules and company-wide finance; separate engagements, contracts and stage instances reference that stable project rather than cloning it.

Domain	Minimum aggregate / entities	Key relationships
Identity & mandates	User, Employee, RoleGrant, ProjectMembership, DelegationMandate, ActingAuthority	User -> granted role/scope; mandatable actions/time-limited deputies.
Client/Project	Client, Enquiry, SiteVisitBooking, Project, ServiceEngagement	One Project many service engagements, visits, modules, client access grants.
Design	SiteSurvey, DesignFileVersion, DisciplineTask, TechnicalReview, ClientApproval, DesignChange	Every approval references exact approved document version.
Commercial	ProposalVersion, Acceptance, ContractVersion, Signature, Variation	Each financial or work gate checks same acceptance/signed contract version.
Compliance	Obligation, Applicability, License, Permit, Certificate, ComplianceHold	Company standard + project applicability and regulator evidence.
People & execution	WorkerProfile, EmployerLink, Assignment, OriginalReport,	Original worker report separate from supervisor verification and PM

Domain	Minimum aggregate / entities	Key relationships
	Verification, Dispute	resolution.
Procurement	Supplier, TrustedContact, BeneficiaryChange, Quote, Evaluation, PO, GRN, Invoice	Approved PO-version -> actual accepted GRN -> supplier invoice -> payment checks.
Stock/assets	StockBatch, Store, StockMovement, Asset, OperatorPermit, DailyInspection, FuelIssue	Custody chain, ownership/title, original worker reports, project costing.
Finance	GeneralLedgerEntry, ProjectCostLedger, Payment, Receipt, Advance, Reconciliation, Journal	No erasure of original financial posts; correcting journals link reversal refs.
Quality/audit	QualityFinding, SafetyHold, AuditEvent, AuditCase, CorrectiveAction, Evidence	Independent audit record append-only and case protection.
Closeout	Handover, SnagItem, Retention, WarrantyTerm, WarrantyClaim	Completion, client acceptance, financial close and warranty expiry separate.

05.1 System-wide state envelope and integrity

- Every workflow action stores id, parent/project/engagement, type, action, actorUserId, representedRole, authorizingMandateId, actionTimestamp, previousState, nextState, reasonCode, attachments, correlationId and source revisions.
- Business state history and evidence attachments must be tamper-evident and non-deletable by ordinary admins. Personally identifying records need legally valid retention and access safeguards; immutable audit does not mean unrestricted public disclosure.
- File versions use content hash, classification, owner, documentPurpose, supersedesVersionId, approvedBy, approvedAt and exact client publication grant. Never expose storage URL without server-side authorization.
- Critical operation endpoints require MFA or step-up verification, replay/idempotency key, server-side authority check and transaction-level segregation of duties. Exceptions reference applicable policy rule.
- Client-facing or worker-facing notifications are async, retriable and logged; original write transaction is authoritative even if a notification fails.

06. Cross-module service/event design

Business event	Consumers / gate
SiteVisitScheduled	Create project entry and site brief task only after valid mutually agreed date.
SiteBriefTechnicallyReleased	Module 2 intake; versioned approved survey and assigned architectural lead.
Client2DApproved	Permit controlled 3D work for that exact design revision.
DisciplinePackageApproved	Release next technical discipline and maintain source references.
ProposalPublished / ProposalAccepted	Client portal accepted version and management verification queue.
PaymentVerified / FinanceExceptionApproved	Re-evaluate eligible financial stage gate without altering receivable balance.
Module4ReleaseAuthorized	Create statutory cases/issue sets per project obligations.
PermitRecorded / ComplianceHoldChanged	Update project readiness and Module 6 mobilization lock.
ContractExecuted / ServiceActivated	Enable only contracted dashboard/tools, planning-only Module 6 as applicable.
MobilizationAuthorized	Permit physical construction tasks only when all nonwaivable preconditions true.
WorkerReportSubmitted	Immediate manager/PM read visibility, separate supervisor verification task, audit.
SupplierBeneficiaryChanged	Flag affected pending payments; force trusted-contact

Business event	Consumers / gate
	revalidation/new authorization.
GRNAccepted / StockMovementVerified	Stock and accounts reconciliation of actual accepted quantities.
EquipmentSafetyRestricted / EquipmentCleared	Immediate targeted asset hold and competent technical documented release.
HandoverAuthorized / WarrantyCaseOpened	Start contract-defined warranty monitoring without wiping financial claims.

07. Security, audit and operational resilience

- Protected audit channel routes allegations involving the GM or a department head to independent auditor and nonconflicted owner/governance decision-maker; access logs also audited.
- HR, auditor and customer data have partitioned confidentiality; least privilege applies at tenant, project, case, record type and individual file-level.
- No transaction creator can approve all stages, no payable payee-change to unverified destination; no duplicate invoices/payroll periods, no hidden versions/stock balance edits.
- Encrypted transport, encryption at rest, segregated key/secrets administration, backups, restore drills, retention schedule and offboarding revocation.
- Offline site/mobile sync must display upload acknowledgment, preserve original local client submission id, prevent duplicates and never label not-yet-synced evidence as completed.
- Alerts escalate based on project risk/severity/threshold; routine site operations continue while only affected unsafe assets/payments/work are restricted.

08. Master dashboard inventory

Workspace	Core widgets / views
Executive / owner	Strategic cash and project portfolio, profitability by project, major contractual/statutory exposure, audit escalations.
General Manager	GM approvals and exceptions, department SLAs, unresolved safety, cash risk, projects exceeding cost/schedule thresholds.
Site/PM command	Site tasks, worker originals vs verification, stock flow, procurement, quality holds, asset utilization, client update queue.
Finance	AR/AP aging, receipts, verified payment batches, escrow-like advances, cash reconciliations, cost codes, payroll.
HR	Role/shift allocation, attendance exceptions, certification expiry, leave/discipline restricted cases.
Compliance / QA	Project obligations, permits/certificates expiry, open stage holds, nonconformity recurrence, corrective verification.
Independent auditor	Risk alerts, immutable records, evidence access logs, protected cases, finding aging and resolution validation.
Client	Personal project service scopes, approved designs, sign-offs, invoices, updates, handover and warranty claims.

09. Build strategy and acceptance

Wave	Objective	Deployment gates
Foundation	Identity and authority, project/client model, versions and immutable audit, document storage.	Prove project isolation, owner vs GM access, delegated authority, backups and security.
M1-3	Lead capture/booking,	Prove exact-version review, client

Wave	Objective	Deployment gates
	2D/3D/engineering workflow, published proposal and finance acceptance.	access, release gates and Accounts reconciliation.
M4-5	Application cases, professional certs, service contracts, signature, compliance holds.	Prove technical signature scope, NCA gate, service flexibility and no construction bypass.
M6 core	Worker portals, daily reports, stores, procurement, QA/safety, accounting commitments.	Prove independent worker evidence, 3-way matching, thresholds and material ownership.
M6 advanced	Equipment, supplier advances, subcontract claims, valuations, variation, handover/warranty.	Prove safety clearance, payee protection, finance forecast and distinct closeout stages.
M7 cross-cutting	Complete HR, audit, finance reporting, compliance, centralized QA and dashboards.	Prove auditor independence, legal restrictions, succession and operational resilience.

Architecture freeze depends on confirmation of actual role holders, corporate signing mandates, approval limits, payroll/legal rules, accounting tax setup, project-specific statutory checklists, notification channels and required integration credentials. These are parameters to obtain, not permission to revisit the already selected business workflow.

10. Developer acceptance criteria (cross-module)

- MAST-UAT-01. Customer intake cannot create active project without agreed site visit date.
- MAST-UAT-02. Only a valid 2D acceptance enables 3D.
- MAST-UAT-03. No engineering handover unless predecessor technically approved.
- MAST-UAT-04. Client may not retrieve working editable CAD/ArchiCAD design files.
- MAST-UAT-05. Client acceptance binds exact published proposal and financial revision.
- MAST-UAT-06. Accounts manually initiates customer overdue reminders.
- MAST-UAT-07. An authorized financial milestone exception does not change receivable ledger.
- MAST-UAT-08. No professional stamp from unverified or expired qualified identity.
- MAST-UAT-09. M4 conditional completion cannot remove construction NCA statutory lock.
- MAST-UAT-10. Client service selection is not an executed contract.
- MAST-UAT-11. Company signature requires approved final version and valid signatory authority.
- MAST-UAT-12. Module 6 can open planning-only while physical execution is blocked.
- MAST-UAT-13. Worker original evidence is instantly visible to PM/GM without supervisor gate.
- MAST-UAT-14. Supervisor disagreement appends review and preserves original media.
- MAST-UAT-15. Unrelated worker cannot see another worker compensation.
- MAST-UAT-16. Supplier bank account change automatically holds affected pending payment.
- MAST-UAT-17. Three-way match against accepted GRN quantity prevents duplicate payment.
- MAST-UAT-18. Stock transit remains in transit until receiving Storekeeper independently accepts.
- MAST-UAT-19. Client-owned stock movement blocked absent ownership entitlement/required client consent.
- MAST-UAT-20. Equipment cannot be assigned to unqualified/expired operator.
- MAST-UAT-21. Critical pre-use defect restricts affected equipment only.
- MAST-UAT-22. Repairer cannot self-certify independent return-to-service safety clearance.
- MAST-UAT-23. Fuel issue has equipment/site/meter/operator/custody evidence.
- MAST-UAT-24. Serious stock losses trigger fair investigation; no automatic salary deduction.
- MAST-UAT-25. A chargeable variation needs management-approved client accepted exact scope.
- MAST-UAT-26. PM cannot publish confidential worker disputes to client portal.
- MAST-UAT-27. Owner cannot bypass GM operational approvals except reserved governance powers.
- MAST-UAT-28. GM-linked whistleblower case bypasses conflicted GM controls.

- MAST-UAT-29. Management financial override never clears safety/technical/permit gate.
- MAST-UAT-30. Physical completion, final account, formal client handover and warranty close remain independently trackable.

11. Source and version register

Controlled file	Content
Module 01 v3	Site Visit - addendum followed by full revised source.
Module 02 v3	Planning - addendum followed by full revised source.
Module 03 v3	Client Acceptance - addendum followed by full revised source.
Module 04 v3	Approvals - addendum followed by full revised source.
Module 05 v3	Contracts - addendum followed by full revised source.
Module 06 v3	Construction - addendum followed by full revised source.
Module 07 v3	Standalone authoritative central HR, Finance, Compliance, QA, Audit, IT and GM governance rule set.
Integration Guide 08 v3	Screen inventory, modules/service interfaces, gates, technical tests and implementation configuration.